BORONG ZHANG

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EDUCATION

University of Wisconsin–Madison — **Madison, WI**
Ph.D. Candidate in Mathematics, Advisors: Prof. Qin Li & Leonardo Andrés Zepeda Núñez — 09/2020 – Present
- **Honors:** Mathematics Department Ascending Scholar Fellowship

University of California, Berkeley — **Berkeley, CA**
Bachelor of Arts in Applied Mathematics & Computer Science — 08/2016 – 06/2020
- **Honors:** High Honors in Applied Mathematics; Distinction in General Scholarship; Dean's Honors Lists

PUBLICATIONS

Zhang, B., Deng, J., Jiang, Y. & Di, Z. W. (2025). Stochastic Multigrid Method for Blind Ptychographic Phase Retrieval. *ArXiv.org*. https://arxiv.org/abs/2511.01793

Zhang, B., Li, Q., & Di, Z. W. (2025). MAGPIE: Multilevel-Adaptive-Guided Solver for Ptychographic Phase Retrieval. *ArXiv.org*. https://arxiv.org/abs/2504.10118

Zhang, B., Guerra, M., Li, Q., & Zepeda-Núñez, L. (2025). Back-Projection Diffusion: Solving the wideband inverse scattering problem with diffusion models. *Computer Methods in Applied Mechanics and Engineering, 443, 118036*. https://doi.org/10.1016/j.cma.2025.118036

Zhang, B., Zepeda-Nunez, L., & Li, Q. (2024). Solving the wide-band inverse scattering problem via equivariant neural networks. *Journal of Computational and Applied Mathematics, 451, 116050–116050*. https://doi.org/10.1016/j.cam.2024.116050

Huang, E. G., Wang, R.-Y., Xie, L., Chang, P., Yao, G., **Zhang, B.**, Ham, D. W., Lin, Y., Blakely, E. A., & Sachs, R. K. (2020). Simulating galactic cosmic ray effects: Synergy modeling of murine tumor prevalence after exposure to two one-ion beams in rapid sequence. *Life Sciences in Space Research, 25, 107–118*. https://doi.org/10.1016/j.lssr.2020.01.001

TALKS

Efficient Symmetry-Driven Diffusion Models for Wideband Inverse Scattering — 10/2025
- SIAM Central States Section Annual Meeting — Fayetteville, AR

Efficient Symmetry-Driven Diffusion Models for Wideband Inverse Scattering (Lightning Talk) — 06/2025
- The Institute for Mathematical and Statistical Innovation — Chicago, IL

Efficient Symmetry-Driven Diffusion Models for Wideband Inverse Scattering — 05/2025
- SIAM Conference on Applications of Dynamical Systems — Denver, CO

Multigrid-based Stochastic Minimization for Ptychographic Phase Retrieval — 04/2025
- Copper Mountain Conference On Iterative and Multigrid Methods — Frisco, CO

Multigrid-based Stochastic Minimization for Ptychographic Phase Retrieval — 03/2025
- Graduate Applied Math Seminar, University of Wisconsin-Madison — Madison, WI

Solving the Inverse Scattering Problem: Leveraging Symmetries for Machine Learning — 11/2024
- SIAM Student Chapter Seminar, University of Wisconsin-Madison — Madison, WI

Solving the Inverse Scattering Problem: Leveraging Symmetries for Diffusion Models — 09/2024
- Graduate Applied Math Seminar, University of Wisconsin-Madison — Madison, WI

CONFERENCES, WORKSHOPS & SUMMER SCHOOLS

SIAM Central States Section Annual Meeting 10/2025
- University of Arkansas — Fayetteville, AR

Statistical and Computational Challenges in SciML 06/2025
- The Institute for Mathematical and Statistical Innovation — Chicago, IL

SIAM Conference on Applications of Dynamical Systems 05/2025
- Sheraton Denver Downtown Hotel — Denver, CO

Copper Mountain Conference On Iterative and Multigrid Methods 04/2025
- Copper Mountain — Frisco, CO

Atlanta SIAM Student Conference 03/2025
- Georgia Institute of Technology — Atlanta, GA

Data-driven PDE-based Inverse Problem, in Theory and Practice 08/2024
- University of Wisconsin-Madison — Madison, WI

Junior Researcher Meeting, on Forward and Inverse Kinetic theory 09/2022
- University of Wisconsin-Madison — Madison, WI

RESEARCH EXPERIENCE & INTERNSHIPS

Givens Associate, MCS Division, Argonne National Laboratory 05/2025 - 08/2025
Supervisor: Dr. Zichao (Wendy) Di — Lemont, IL
- Proposed a stochastic multigrid method for blind ptychographic phase retrieval that employs a geometric-mean, phase-aligned joint object–probe update, with convergence guarantees.
- Designed, implemented, and benchmarked the algorithm in Python (NumPy/PyTorch).

Research Aide, MCS Division, Argonne National Laboratory 06/2024 - 09/2024
Supervisor: Dr. Zichao (Wendy) Di — Lemont, IL
- Proposed a stochastic multigrid, majorization–minimization framework for ptychographic phase retrieval.
- Delivered a comprehensive theoretical analysis of convergence.
- Designed, implemented, and benchmarked the algorithm in Python.

TEACHING EXPERIENCE

Math 340: Elementary Matrix and Linear Algebra, TA	Fall 2025
Math 211: Calculus, Head TA	Spring 2025
Math 112: College Algebra, Instructor	Fall 2023-Fall 2024
Math 234: Calculus - Functions of Several Variables, TA	Spring 2023
Math 211: Calculus, TA	Fall 2022
Math 521: Analysis I, TA	Summer 2022
Math 222: Calculus and Analytic Geometry II, TA	Spring 2021
Math 221: Calculus and Analytic Geometry I, TA	Fall 2020

ORGANIZATIONS & OUTREACH

Graduate Peer Mentor Fall 2025

Directed Reading Summer 2025-Fall 2025
- Topic: Stochastic Multigrid Methods for Ptychographic Phase Retrieval

Directed Reading Program, Mentor Spring 2025
- Topic: Stochastic Differential Equations: Score-Based Diffusion Models

Directed Reading Program, Mentor Fall 2024
- Topic: Solving the Inverse Scattering Problem: Classical Methods and Machine Learning

SKILLS & INTERESTS

Programming Languages: Python, Java, MatLAB, Julia, R, C
Libraries, APIs, and Technologies: Git, Jupyter, SciPy, NumPy, JAX, Flax, Tensorflow, PyTorch